DIMENSION STRATIFICATION OF κ_{ch} ON THE CALABI–YAU LANDSCAPE

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Abstract. For compact Calabi–Yau d-folds, the chiral-shadow invariant $\kappa_{\mathrm{ch}}(\mathcal{A}_X)$ equals the Hodge supertrace $\sum_q (-1)^q h^{0,q}(X)$ (`thm:kappa-hodge-supertrace-identification`) unconditionally via HKR plus the Mukai pairing plus an HC_d^- trace. The dimension stratification $d \in \{1, 2, 3, 4, 5\}$ then reads off explicitly (`thm:kappa-stratification-by-d`): elliptic curves contribute 0; K3 contributes 2; abelian and bielliptic contribute 0; quintic, $K3 \times E$, $E^{\times 3}$ contribute 0; local $\mathbb{P}^2$ contributes $3/2$ (`thm:local-p2-shadow`); CY_4 sextic contributes 2; CY_5 generic 0. The conifold has resolved form $\mathrm{Tot}(\mathcal{O}(-1)^2 \to \mathbb{P}^1)$, so it lies outside the local-surface family $\mathrm{Tot}(K_S \to S)$. The direct McKay route gives $\kappa_{\mathrm{ch}}(\text{conifold}) = 1$.

Primary theorems

- `thm:kappa-hodge-supertrace-identification` (unconditional, compact CY, via HKR + Mukai + HC_d^-).
- `thm:kappa-stratification-by-d` (explicit tabulation $d = 1..5$).
- `thm:local-p2-shadow` $\kappa_{\mathrm{ch}} = 3/2$.
- `cor:conifold-non-local-surface` (resolves the conifold-vs-local-surface classification).
- `thm:borcherds-weight-kappa-BKM-universal` ($\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ universal across $N \in \{1, 2, 3, 4, 6\}$; $N = 1$ coincidence with $\kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ fails for $N \geq 2$, resolving the naive $N = 1$ identification).

Chapter inputs

```
\input{../chapters/examples/cy_d_kappa_stratification.tex}
\input{../chapters/examples/derived_categories_cy.tex}
\input{../chapters/examples/matrix_factorizations.tex}
\input{../chapters/examples/k3e_cy3_programme.tex}
```

Cross-volume dependencies

- Vol I landscape_census.tex (canonical κ_{ch} formulas).
- Vol II `\ref{part:ht-landscape}` (HT landscape; compare κ_{ch} on HT-gauge side).

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